CSA4707-DEEP LEARNING

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Experiment 1: Vector and Matrix Operations for Deep Learning

Problem Statement: An e-commerce firm represents customers and products as numeric vectors to power its recommendation engine. Every recommendation is a chain of dot products and matrix multiplications. A small online bookstore has 6 customers and 5 genres. Each customer has a rating vector, and each book has a genre-weight vector. The store wants to score every customer–book pair and find similar customers. The objective is to perform vector and matrix operations that underpin a neural network layer. Use NumPy, Matplotlib.

Tasks:

1. Build the customer–genre (6×5) and genre–book (5×4) matrices in NumPy.

2. Compute dot products, matrix product, transpose, inverse (where valid), and element-wise operations.

3. Compute cosine similarity between customers and identify the two most similar.

4. Show that a dense layer is y = Wx + b by computing it manually for one customer.

5. Visualise the score matrix as a heatmap.

FORMULA:

1. Matrix multiplication

$$S = A \times B$$

2.Dot product

$$a \cdot b = \sum_{i=1}^{n} a_i b_i$$

3.Dense layer

$$y = Wx + b$$

4.Cosine Similarity = (A · B) / (||A|| * ||B||)

5. Transpose and inverse

$$A^T = TransposeofA$$

$$M^{-1}M = I$$

PROGRAM CODE:

```
import numpy as np
import matplotlib.pyplot as plt

A = np.array([
    [5, 4, 1, 2, 3],
    [4, 5, 1, 2, 3],
    [1, 2, 5, 4, 2],
    [2, 1, 4, 5, 3],
    [5, 3, 2, 1, 4],
    [1, 2, 4, 5, 1]
])

B = np.array([
    [0.9, 0.2, 0.1, 0.3],
    [0.8, 0.3, 0.2, 0.1],
    [0.1, 0.9, 0.8, 0.2],
    [0.2, 0.7, 0.9, 0.4],
    [0.3, 0.2, 0.3, 0.9]
])

print("Dot Product:", np.dot(A[0], A[1]))

S = A @ B
print("Score Matrix:\n", S)

print("Transpose:\n", A.T)

M = A.T @ A + np.eye(5)
print("Inverse:\n", np.linalg.inv(M))
```

```
print("Addition:\n", A + A)
print("Multiplication:\n", A * A)

C = (A @ A.T) / np.outer(
    np.linalg.norm(A, axis=1),
    np.linalg.norm(A, axis=1)
)
print("Cosine Similarity:\n", np.round(C, 2))

C[range(6), range(6)] = 0
i, j = np.unravel_index(np.argmax(C), C.shape)
print("Most Similar Customers:", i + 1, "and", j + 1)

x = A[0]
W = B.T
b = np.ones(4)
y = W @ x + b
print("Dense Layer Output:", y)

plt.imshow(S, cmap="Blues")
plt.colorbar()
plt.xlabel("Books")
plt.ylabel("Customers")
plt.title("Recommendation Score Heatmap")
plt.show()
```

OUTPUT:

```
Dot Product: 54
Recommendation Scores:
 [[9.1 5.1 4.8 5.6]
 [9.  5.2 4.9 5.4]
 [4.4 8.5 8.7 4.9]
 [4.9 8.4 9.  6.2]
 [8.5 5.2 4.8 6.2]
 [4.2 8.1 8.5 4.2]]
Transpose:
 [[5 4 1 2 5 1]
 [4 5 2 1 3 2]
 [1 1 5 4 2 4]
 [2 2 4 5 1 5]
 [3 3 2 3 4 1]]
Inverse:
 [[ 0.27381918 -0.11888915  0.06749737  0.00945782 -0.260066  ]
 [-0.11888915  0.1374435   0.00901624 -0.03781121  0.02782751]
 [ 0.06749737  0.00901624  0.18646681 -0.13861569 -0.11371866]
 [ 0.00945782 -0.03781121 -0.13861569  0.14475565  0.00940162]
 [-0.260066    0.02782751 -0.11371866  0.00940162  0.37475613]]
Addition:
 [[10  8  2  4  6]
 [ 8 10  2  4  6]
 [ 2  4 10  8  4]
 [ 4  2  8 10  6]
 [10  6  4  2  8]
 [ 2  4  8 10  2]]
Element-wise Multiplication:
 [[25 16  1  4  9]
```

[16 25 1 4 9]

[1 4 25 16 4]

[4 1 16 25 9]

[25 9 4 1 16]

[1 4 16 25 1]]

Cosine Similarity:

[[1.   0.982 0.61 0.673 0.964 0.59]

[0.982 1.   0.629 0.655 0.927 0.61]

[0.61 0.629 1.   0.953 0.629 0.97]

[0.673 0.655 0.953 1.   0.691 0.944]

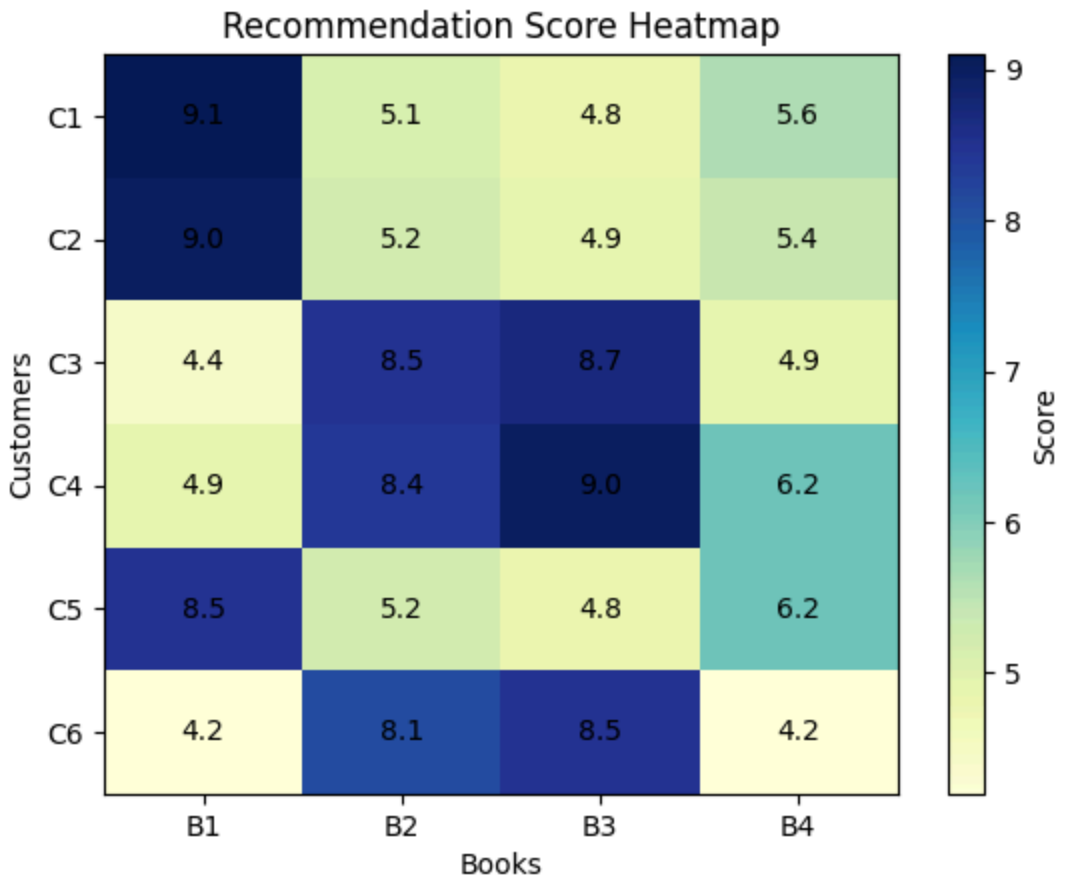
[0.964 0.927 0.629 0.691 1.   0.551]

[0.59 0.61 0.97 0.944 0.551 1.   ]]

Most Similar Customers: 1 and 2

Similarity Score: 0.982

Dense Layer Output: [10.1 6.1 5.8 6.6]


Recommendation Score Heatmap
C1
C2
C3
C4
C5
C6
Customers
9.1
5.1
4.8
5.6
9.0
5.2
4.9
5.4
4.4
8.5
8.7
4.9
4.9
8.4
9.0
6.2
8.5
5.2
4.8
6.2
4.2
8.1
8.5
4.2
B1
B2
B3
B4
Books
9
8
7
6
5
Score